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New releases focus on higher bit rates

Due to an increase in broadband services supporting 4K/8K video streaming, there has been an expansion in optical access networks using passive optical network technology. These are rapidly moving towards higher speeds, while moving to newer standards at the same time, thus, the development of optical line terminal and optical network units for passive optical network systems that require wideband performance and precise measurement.

MP1800A from Anritsu, for example, is a plug-in modular bit error rate tester for measuring multichannel interfaces up to 32Gbps. MX180014A software controls MP1800A to generate two-channel test-signal burst patterns and set skew. OLT evaluation input sensitivity and timing tests can be performed using the GUI by setting test signal pattern length and timing.

R&S TS-5GCS test setup combines channel-sounding software with the signal and spectrum analyser and the vector signal generator. The setup supports development of applications requiring multichannel scenarios up to 40GHz, allowing measurement of channels in 5G frequency bands in centimetre and millimetre ranges.

Very recently, Huwin has introduced a new USB 3.1 Type C test solution. The test fixtures have been designed to measure USB 3.1 Type C connectors using a commercial network analyser.

Analysing future networks

Over the past couple of decades, VNAs have increased in popularity over scalar network analysers due to their capability to measure magnitude and phase-over wide-frequency sweeps. The latest generation of VNAs can also measure non-linear behaviour of active devices, like gain and phase compression, IMD, AM-AM and AM-PM conversion, and spurious response.

Advanced technologies like automotive RADAR, 5G and Wi-Gig, and applications like THz imaging and material measurements at millimetre-wave, mandates designers and researchers to characterise components beyond 67GHz. Add that to the ever-growing list of features in instruments and we come up with the possibility that network analysers may not even be recognisable soon. EFY